

around 650 Internet-providing LEO satellites at 800km and 950km, by 2018. Its satellites are designed with lesser components and with a lighter weight, so these can be mass manufactured and also launched easily.

Another notable aspect is their patent-pending Progressive Pitch technology, which enabled them to gain use of the priority spectrum allotted by United Nations International Telecommunications Union in 1997. In order to use the spectrum, OneWeb had to work out the challenge of not interfering with GEO satellites. Their Progressive Pitch technology helps them do this by gradually and slightly tilting their satellites as they approach the Equator to make sure they never cause, or receive, interference.

Dynamic spectrum sharing (DSA) is also becoming a mature field since 4G LTE introduced priority based structures both for control signals and user traffic.

Airbus Group's infrastructure would be used for manufacturing the satellites, while Arianespace will help in deploying these on time. Coca Cola will set up millions of access points across the Earth.

OneWeb's architecture will provide layer-2 and layer-3 services, which any telecom company can use to extend its network. Users can connect from anywhere in the world using a small and affordable, solar- or battery-powered OneWeb device. Thanks to Bharti's involvement, this technology promises to wave its magic wand over India, too.

O3b Networks. A lesser-known player, O3b Networks' motto is 'fibre speed, satellite reach.' The Google-backed company serves customers in around 180 countries with low-cost, high-speed, low-latency Internet and mobile connectivity with a scalable constellation of 12 MEO satellites, designed, integrated and tested by Thales Alenia Space.

One of its divisions, O3b Mari-

Light to the rescue...

Some years ago, Dr Harald Haas of University of Edinburgh proposed that LED bulbs could be used to transmit data. Li-Fi or light fidelity is a wireless communications technology like Wi-Fi, but it uses light from LED bulbs instead of radio waves to transmit data. A chip fitted in the LED bulb makes it flicker on and off thousands of times per second to communicate the zeroes and ones that make our data. The light is switched on and off so fast that it is not visible to the human eye. Current demonstrations of Li-Fi show that it is 100 times faster than Wi-Fi, and Haas believes it has the potential to achieve data densities 1000 times greater than Wi-Fi. His company is ready to commercialise the technology in the near future.

Velmenni, an Estonian start-up also revealed recently that they have started trialling Li-Fi in offices and industrial environments in Tallinn. Last year, NASA, too, started researching on how Li-Fi can be used in outer space. According to NASA, "One of the primary advantages of visual light communications (VLC) over radio-frequency bandwidth limitations is that the visible light spectrum is 10,000 times larger. A VLC network provides a wireless or fibre-less light-photon medium with virtually unlimited data transfer and significantly-reduced security risks. It also operates with reduced energy requirements, thus making it a green technology."

Li-Fi sounds interesting to trend-watchers because it could free up the crowded radio spectrum. Dr Haas also claims that it could help connect remote locations using transmitters and solar panels installed on hilltops and rooftops. However, some experts feel that it still has a long way to go as its efficiency is reduced when light is blocked, say, under foggy conditions. Once these problems are ironed out, Li-Fi does promise to reveal a whole new connected world, perhaps in the next decade.

time, has been attracting attention of late, with its interesting offering for improving connectivity on ships. Although satellites have long been used for mobile connectivity on ships, these have been problematic because there are black spots on oceans, which are not covered. Plus, when the ship changes direction, it takes some time to link up with the satellite network again, all of which tends to irritate customers with frequently-interrupted connectivity.

To avoid this, O3b's satellite beams follow the ship on its normal route, so that the ship is always maintained within the beam centre. O3b receives latitude/longitude updates on two-hour intervals via in-band or out-of-band channels, and beam-tracking updates in real time if the ship has to change course.

Inmarsat. An early example in this space, Inmarsat serves a limited set of customers for high-bandwidth applications, using LEO satellites. It is now readying its next-generation constellation, Global Xpress, which promises to remove the black spots in ship-to-shore communications, and ensure reliable, high-speed

broadband connectivity to ships.

Inmarsat is working with Rolls Royce on their autonomous, cargo-carrying drone ship project, and with Ericsson to facilitate sharing of cargo, logistics and vessel-operation-al data to help streamline the maritime supply chain. You could think of this as something like a maritime extension of the Internet of Things.

Mobile might continue to rule

"The telecommunications landscape is replete with limited successes of several innovative satellite based communication systems including Iridium, Teledesic and Globalstar. It may be noted that consistent with market needs and technologies available at that time, their goals were comparatively modest—providing basic voice capabilities to customers. The business model for these initiatives broke down since competing mobile systems provided lower-cost alternatives, albeit allowing selective coverage instead of universal coverage.

"The claim of current satellite based initiatives like SpaceX and OneWeb—that cost structures have